Troy James Larsen

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Education

NC State University *Raleigh, NC*

PH.D. IN APPLIED MATHEMATICS *2025 - Pres.*

- Advisor: Dr. Alen Alexanderian
- Preparing the Professoriate (2026-2027)
- Advanced Teaching Certificate (July 2025)
- GPA: 4.00/4.00

M.S. IN APPLIED MATHEMATICS *2023 - 2025*

- Level 1 Teaching Certificate (May 2025)
- GPA: 4.00/4.00

Budapest Semesters in Mathematics *Budapest, Hungary*

POST-BACCALAUREATE STUDY *2022 - 2023*

- High Honors (May 2023)

Washington & Lee University *Lexington, VA*

B.S. IN MATHEMATICS AND CLASSICS *2018 - 2022*

- Honors Thesis Advisor: Dr. Aaron Abrams
- Phi Beta Kappa, *summa cum laude*
- GPA: 3.99/4.00

Professional Experience

2026-Pres.	**Graduate Teaching Assistant**, NC State University
2026	**Young Scholar**, Comparative Medicine Institute, NC State University
2026	**Graduate Student Mentor**, DRUMS REU, NC State University
2025-2026	**National Science Foundation RTG Fellow**, NC State University
2025	**Graduate Student Mentor**, DRUMS REU, NC State University
2023-2025	**Graduate Teaching Assistant**, NC State University
2022-2023	**Fulbright Scholar**, Budapest Semesters in Mathematics
2021-2022	**Undergraduate Research Assistant**, Washington & Lee University
2021	**National Science Foundation REU Student**, University of Virginia
2020	**Summer Research Scholar**, Washington & Lee University

Research Interests

General: Uncertainty quantification, numerical analysis and scientific computing, inverse problems, applied probability

Specific: Sensitivity analysis, reproducing kernel Hilbert spaces, kernel-based dependence measures, surrogate models

Graduate Coursework

Qualifying Exams: Applied Matrix Theory I/II, Functional Analysis, Measure Theory, Partial Differential Equations I/II

Additional Courses: Advanced Functional Analysis, Computational Inverse Problems, Numerical Analysis I/II, Probability & Stochastic Processes I/II, Uncertainty Quantification

Publications

Journal Articles:

E. Denne, **T. Larsen** (2023). Linking Number and Folded Ribbon Unknots. *J. Knot Theory Ramif.* 32(1).

E. Denne, J. C. Haden, **T. Larsen**, E. Meehan (2022). Ribbonlength of Families of Folded Ribbon Knots. *Involve* 15(4), 591–628.

Unpublished/In Progress:

T. Larsen, A. Alexanderian. *A new kernel-based index for the global sensitivity analysis of models with correlated inputs*. In review, SIAM/ASA Journal of Uncertainty Quantification.

T. Larsen, A. Alexanderian. *Kernel tuning for the HSIC-based GSA of high-dimensional models*. In preparation.

A. Alexanderian, S. Ghosh, J. C. Hall, **T. Larsen**, M. S. Olufsen, C. Toumbleston, D. Upadhyaya. *HSIC-Informed Model Reduction and Selection Applied to a Model of the Valsalva Maneuver*. In preparation.

C. Haggard, J. Introne, Z. Lauko, J. Sapozhnikov, **T. Larsen**, A. Alexanderian, J. Hart. *Optimal path continuation for PDE-constrained inverse problems*. In preparation.

Theses/Reports:

T. Larsen (2022). *Honors Thesis*: On the Realizability of Projective Configurations.

K. Betts, **T. Larsen**, J. Utley, A. Vanis (2021). *REU Report*: On the Tri-Pants Graph, arXiv:2111.07136.

Awards, Fellowships, and Grants

2026	**Graduate Student Leadership Award**, College of Sciences, NCSU	*$ 500*
	Young Scholars Research/Mentorship Grant, Comparative Medicine Institute, NCSU	*$ 9,000*
	National Science Foundation RTG Fellowship, NSF Grant DMS-2342344: UQ4Life	–
	Franke-Norris-Griggs Award for Excellence in Service, NCSU Dept. of Mathematics	*$ 1,000*
2025	**National Science Foundation RTG Fellowship**, NSF Grant DMS-2342344: UQ4Life	–
	Most Outstanding Teaching Assistant Award, NCSU Graduate School	*$ 500*
	Maltbie Teaching Award, NCSU Dept. of Mathematics	*$ 1,000*
	Franke-Norris-Griggs Award for Excellence in Teaching, NCSU Dept. of Mathematics	*$ 1,000*
2023	**Goodnight Doctoral Fellowship**, NCSU Graduate School	*$ 40,000*
	University Graduate Fellowship, NCSU College of Sciences	*$ 4,000*
2022	**Fulbright Grant: Study/Research, Hungary**, BSM/Rényi Institute	–
	Rhodes Scholarship Finalist, District 4	–
	Henry Ruffner Scholarship, W&L Dept. of Academic Affairs	*$500*
	Frank J. Gilliam Award, W&L Dept. of Student Affairs	*$1,000*
	Leader of the Year Award, ODK Alpha Circle	*$100*
	Williams Prize, W&L Dept. of Mathematics	*$500*
	Matthew Mason Scholarship in Latin, W&L Dept. of Classics	*$500*
2021	**Barry Goldwater Scholarship**, Mathematical Sciences	*$7,500*
	Rhodes Scholarship Finalist, District 9	–
	James McDowell Scholarship, W&L Dept. of Academic Affairs	*$1,000*
	Madison Montgomery Shinaberry '16 Award, W&L Dept. of Student Affairs	*$500*
	Taylor Prize, W&L Dept. of Mathematics	*$500*
2020	**Summer Research Scholars Grant**, W&L Dept. of Mathematics	*$5,000*
	Most Outstanding Peer Counselor Award, W&L Dept. of Counseling Services	*$250*

2018	**Johnson Scholarship**, W&L Johnson Program	*$300,000*
	Good Citizenship Award, Philadelphia Union League	*$500*

Presentations and Talks

T. Larsen. 2027. *Kernel Tuning for HSIC-Based Global Sensitivity Analysis of High-Dimensional Models*. SIAM Conference on Computational Science and Engineering, Pittsburgh, PA.

A. Alexanderian, **T. Larsen**. 2026. *Total HSIC: A New Index for Global Sensitivity Analysis* (Poster). SIAM Conference on Uncertainty Quantification, Minneapolis, MN.

A. Alexanderian, **T. Larsen**. 2026. *Total HSIC: A New Index for Global Sensitivity Analysis* (Poster). ICERM Workshop on Bayesian Inverse Problems and UQ, Brown University, Providence, RI.

T. Larsen. 2026. *Introducing the Total HSIC Index for Global Sensitivity Analysis.* Applied Mathematics Graduate Student Seminar, NC State University, Raleigh, NC.

T. Larsen. 2026. *Introducing the Total HSIC Index for Global Sensitivity Analysis.* UQ4Life Seminar, NC State University, Raleigh, NC.

T. Larsen. 2025. *Theory and Applications of Reproducing Kernel Hilbert Spaces.* Applied Mathematics Graduate Student Seminar, NC State University, Raleigh, NC.

T. Larsen. 2025. *Kernel-Based Methods for Global Sensitivity Analysis.* UQ4Life Seminar, NC State University, Raleigh, NC.

J. C. Hall, **T. Larsen**. 2025. *Time-Series Analysis for the Life Sciences.* Think, Collaborate & Do Research Summit, Comparative Medicine Institute, NC State University, Raleigh, NC.

T. Larsen. 2025. *An Introduction to Global Sensitivity Analysis in Weighted Spaces.* Applied Mathematics Graduate Student Seminar, NC State University, Raleigh, NC.

T. Larsen. 2022 *On the Realizability of Projective Configurations*. Honors Thesis Symposium, Washington & Lee University, Lexington, VA.

K. Betts, **T. Larsen**, J. Utley, A. Vanis. 2021. *The Tri-Pants Graph.* Virginia Topology Conference, University of Virginia, Charlottesville, VA.

T. Larsen. 2021. *An Algebraic Approach to Sloane's Persistence Problem* (Poster). Science, Society, and the Arts, Washington & Lee University, Lexington, VA.

J. C. Haden, **T. Larsen**. 2020. *Folded Ribbon Knots and Ribbonlength.* SUMS Conference, James Madison University, virtual.

T. Larsen. 2020. *Mathematical Literacy in the Early Roman Empire* (Poster). North American Congress of Greek and Latin Epigraphy, Georgetown University, Washington, D.C.

Teaching Experience

Win. 2027	**MA 225: Foundations of Advanced Mathematics**, Instructor of Record, NCSU	*Students TBD*
Fall 2026	**MA 141: Calculus I (Honors)**, Recitation Leader, NCSU	*42 students*
Fall 2024	**MA 131: Calculus for Life Sciences & Management A**, Instructor of Record, NCSU	*93 students*
Sum. 2024	**MA 107: Precalculus I**, Instructor of Record, NCSU	*7 students*
Win. 2024	**MA 241: Calculus II**, Recitation Leader, NCSU	*96 Students*
Fall 2023	**MA 141: Calculus I**, Recitation Leader, NCSU	*94 Students*
Fall 2022	**SAT Preparation for Hungarian High School Students**, Presenter, EducationUSA	*47 students*
Fall 2021	**MATH 101: Calculus I**, Course Assistant, W&L	*55 students*
Fall 2021	**FYE 100: General Success**, Course Facilitator, W&L	*17 students*
Fall 2020	**FYE 100: General Success**, Course Facilitator, W&L	*15 students*
Fall 2019	**FYE 100: General Success**, Course Facilitator, W&L	*19 students*

Mentoring

Fall 2026	**Directed Reading Project**, Sensitivity Analysis with Correlated Inputs, NCSU	*2 students*
Win. 2026	**Directed Reading Project**, Binary Classification with Kernel Methods, NCSU	*3 students*
Sum. 2026	**DRUMS REU Project**, Computational Methods for Parameterized Inverse Problems, NCSU	*4 students*
Sum. 2026	**YSP Project**, A Valsalva Maneuver Model to Phenotype Autonomic Dysfunction, NCSU	*1 student*
Sum. 2026	**DRUMS REU Project**, Biomarkers of Dysfunction in a Model of the Valsalva Maneuver, NCSU	*2 students*
2024-Pres.	**Undergraduate Academic Mentoring**, Undergrads Union Grads, NCSU	*5 students*
2021-2022	**Johnson Scholars Mentoring**, Johnson Program, W&L	*2 students*

University and Departmental Service

Leadership Experience:

2027	**SIAM CSE**, Minisymposium Organizer: *Modern Methods in Global Sensitivity Analysis*	*NCSU*
2026-Pres.	**Applied Mathematics Graduate Student Seminar**, Seminar Chair	*NCSU*
2026	**UQ4Life RTG**, Seminar Chair	*NCSU*
2026-Pres.	**UUG Directed Reading Program**, Founding Organizer	*NCSU*
2025-Pres.	**Triangle Area Spectra Chapter**, Secretary	*NCSU*
2024-Pres.	**Undergraduates Union Graduates (UUG)**, President	*NCSU*
2024-Pres.	**Society for Undergraduate Mathematics Seminar Series**, Organizer	*NCSU*
2021-2022	**Phi Eta Sigma First-Year Honors Society**, Chapter President	*W&L*

Committee Work:

2024-Pres.	**Departmental Faculty, Staff, and Student Success Committee**, Student Representative	*NCSU*
2022-2024	**Board of Trustees Residential & Social Life Working Group**, Student Representative	*W&L*
2021-2022	**University Diversity, Equity, and Inclusion Committee**, Student Representative	*W&L*
2021-2022	**Curriculum Subcommittee: Student Leadership and DEI**, Student Representative	*W&L*
2019-2022	**First-Year Experience Committee**, Founding Member	*W&L*
2019-2022	**Dean for Inclusion and Engagement's Student Advisory Board**, Founding Member	*W&L*
2019-2022	**FYE 100 Development and Assessments Committee**, Founding Member	*W&L*
2020-2021	**Student Advisory Board to the Board of Trustees**, Committee Member	*W&L*
2020-2021	**Office of Inclusion and Engagement Assessments Subcommittee**, Student Representative	*W&L*
2020-2021	**Faculty General Education Committee**, Student Advisor	*W&L*

Outreach:

2026	**Graduate School New TA Workshop**, Experienced TA Panelist	*NCSU*
2026	**Dept. of Mathematics Graduate Orientation**, Student Life & Success Panelist	*NCSU*
2026	**Reach for the Stars STEM Fair**, Dept. of Mathematics Representative	*NCSU*
2026	**NC Science Olympiad STEM Expo**, Dept. of Mathematics Representative	*NCSU*
2026	**Dept. of Mathematics Graduate Recruitment**, Research Spotlight Speaker	*NCSU*
2026	**Dept. of Mathematics Graduate Recruitment**, Student Organizations Panelist	*NCSU*
2025	**University Open House**, Dept. of Mathematics Representative	*NCSU*
2025	**Wolfpack STEM Tailgate**, Dept. of Mathematics Representative	*NCSU*
2025	**Reach for the Stars STEM Fair**, Dept. of Mathematics Representative	*NCSU*
2025	**Dept. of Mathematics Graduate Orientation**, Student Organizations Panelist	*NCSU*
2025	**Dept. of Mathematics Graduate Recruitment**, Student Organizations Panelist	*NCSU*
2025	**Dept. of Mathematics Graduate Orientation**, Student Life & Success Panelist	*NCSU*

Search and Hiring Committees:

2021	**Provost**, Interviewer	*W&L*
2021	**Dean of the College**, Committee Member	*W&L*
2021	**Associate Provost for Diversity, Equity, and Inclusion**, Committee Member	*W&L*
2021	**Assistant Professor of Mathematics**, Interviewer	*W&L*
2021	**Visiting Assistant Professor of Mathematics**, Interviewer	*W&L*
2020	**Assistant Director of Diversity and Inclusion**, Committee Member	*W&L*
2020	**Assistant Director of LGBTQ+ Support**, Committee Member	*W&L*
2020	**Johnson Program Coordinator**, Interviewer	*W&L*